



Unsupervised Data Mining

BSCS 3A

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OUTLINE

- Clustering
- Topic Modeling



PRE-ACTIVITY: PRE-ACTIVITY: “Group Them Without Rules”

Situation:

You are given a list of customer transactions from a food ordering system. However, there are:

✗ No labels

✗ No categories

✗ No instructions on grouping

Your task is to **organize the data into meaningful groups based on your own reasoning.**

Transaction ID	Items Ordered
T1	Chicken, Rice, Coke
T2	Burger, Fries
T3	Sundae, Pie
T4	Chicken, Rice
T5	Fries, Coke
T6	Burger, Coke
T7	Pie, Sundae
T8	Chicken, Rice, Juice

✓ Analyze the transactions

✓ Group them into clusters

✓ Give each cluster a **name**

Explain:

Why did you group them this way?

What similarities did you observe?

PRE-ACTIVITY: “Group Them Without Rules”

- Cluster 1: Full Meals
(Chicken, Rice, Drink)
- Cluster 2: Snacks
(Burger, Fries)
- Cluster 3: Desserts
(Sundae, Pie)
- Cluster 4: “Messy Data” - Neutral

PRE-ACTIVITY: “Group Them Without Rules”

“Did I tell you how many groups to make?”

“Did I give you labels?”

“Why are your answers different?”

“What you just did is called **Clustering**.”

CLUSTERING IN DATA MINING

What is Clustering?

Clustering is an **unsupervised learning technique** used to group data points such that:

- Objects in the **same cluster are similar**
- Objects in **different clusters are dissimilar**

Unlike classification, clustering:

- Does **not use labeled data**
- Discovers **hidden structures or patterns**

CLUSTERING IN DATA MINING

Objectives of Clustering

1. Identify **natural groupings** in data
2. Simplify large datasets
3. Support **decision-making**
4. Reveal **unknown patterns**

CLUSTERING IN DATA MINING

Key Concepts

- **Cluster.** A collection of similar data points
- **Similarity / Distance Measures.** Used to determine how close data points are:
 - ✓ **Euclidean Distance** (continuous data)
 - ✓ **Manhattan Distance**
 - ✓ **Cosine Similarity** (text data)

Centroid. The “center” of a cluster (used in K-Means)

Types of Clustering Techniques

1. Partition-Based Clustering (K-Means)

- Divides dataset into **K predefined clusters**

Process

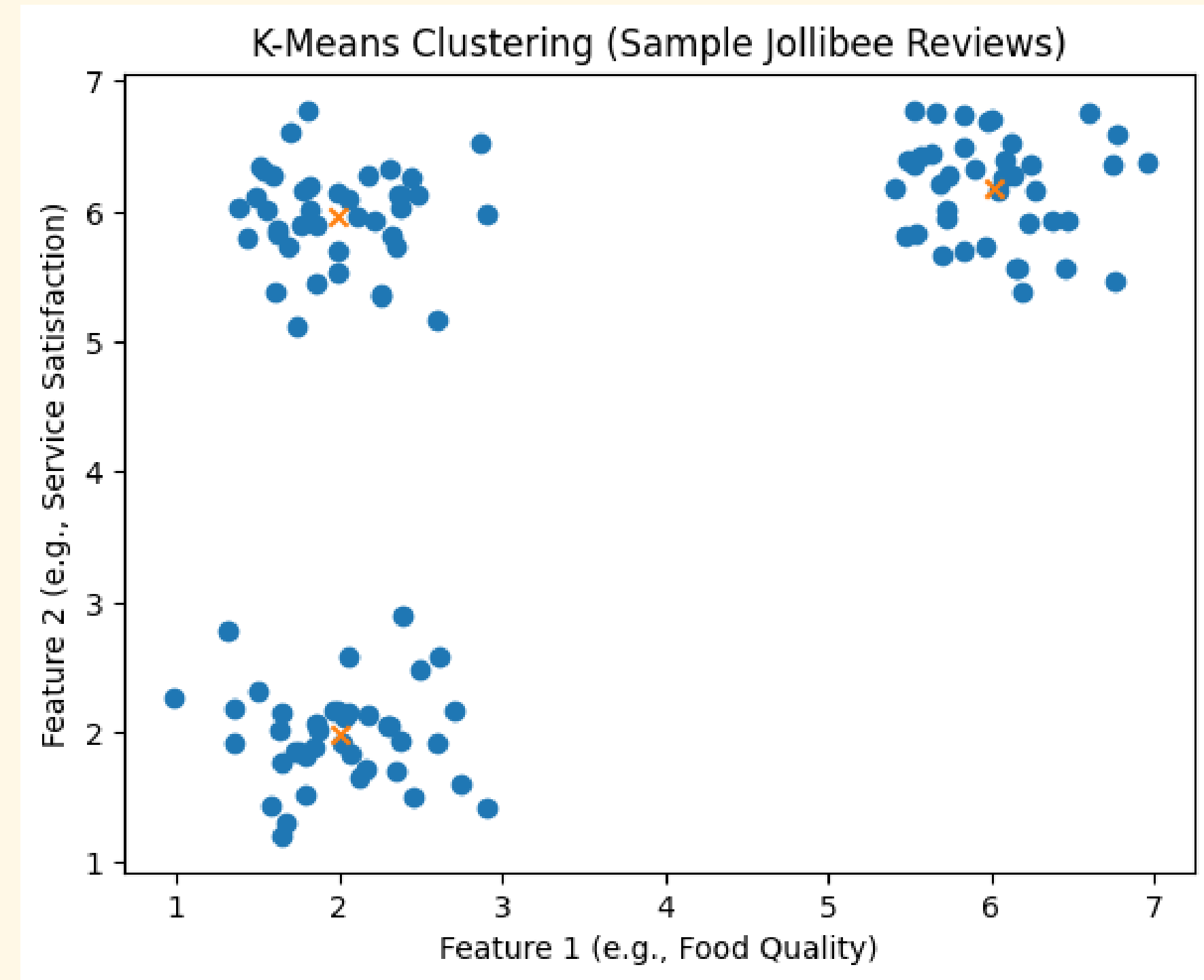
1. Select number of clusters (K)
2. Initialize centroids
3. Assign data points to nearest centroid
4. Update centroids
5. Repeat until convergence

Strengths

- Simple and efficient
- Works well for large datasets

Limitations

- Must choose K beforehand
- Sensitive to outliers



Types of Clustering Techniques

2. Hierarchical Clustering

- Builds a **tree of clusters** (dendrogram)

Types

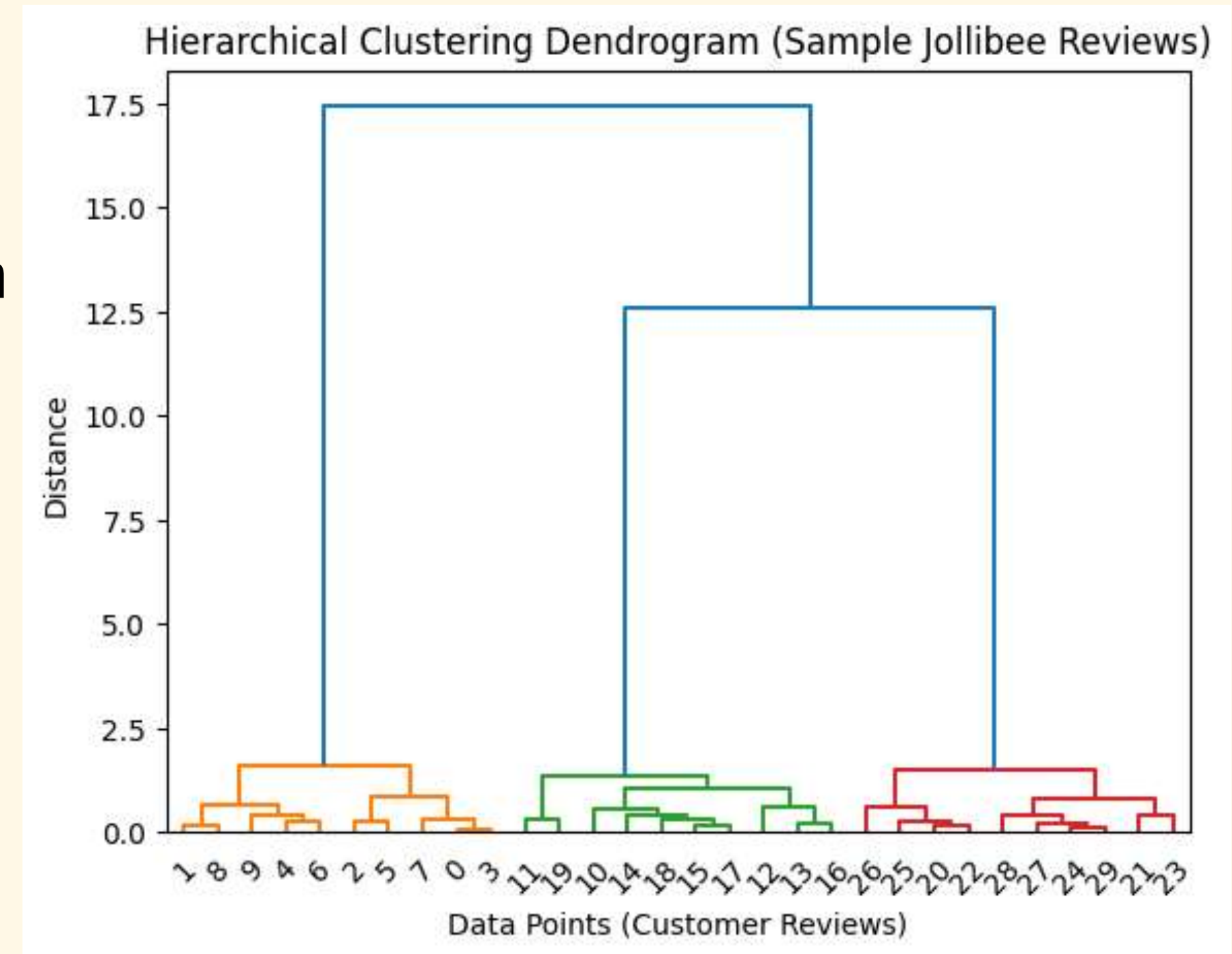
- ✓ **Agglomerative (bottom-up)**: start with individual points, merge
- ✓ **Divisive (top-down)**: start with one cluster, split

Strengths

- ✓ No need to specify number of clusters initially
- ✓ Provides hierarchy of clusters

Limitations

- ✓ Computationally expensive
- ✓ Difficult to scale



Types of Clustering Techniques

3. Density-Based Clustering (DBSCAN)

- Clusters are formed based on **dense regions**

Key Idea

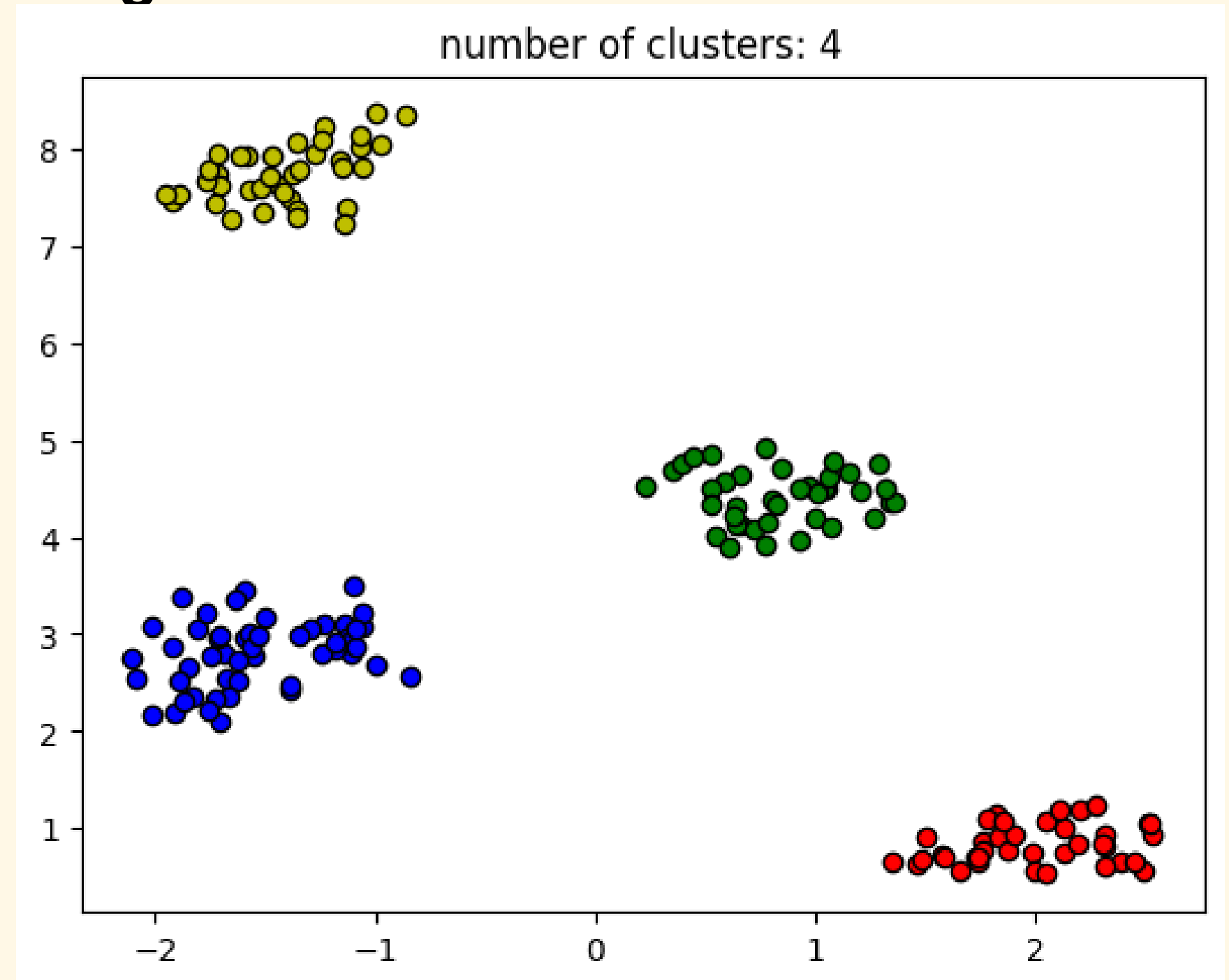
- ✓ Points in dense regions → cluster
- ✓ Sparse points → noise/outliers

Strengths

- ✓ Detects arbitrary shapes
- ✓ Handles noise well

Limitations

- ✓ Sensitive to parameter selection
- ✓ Struggles with varying densities



Types of Clustering Techniques

4. Model-Based Clustering

- Assumes data is generated from a mixture of **probability distributions**

Example

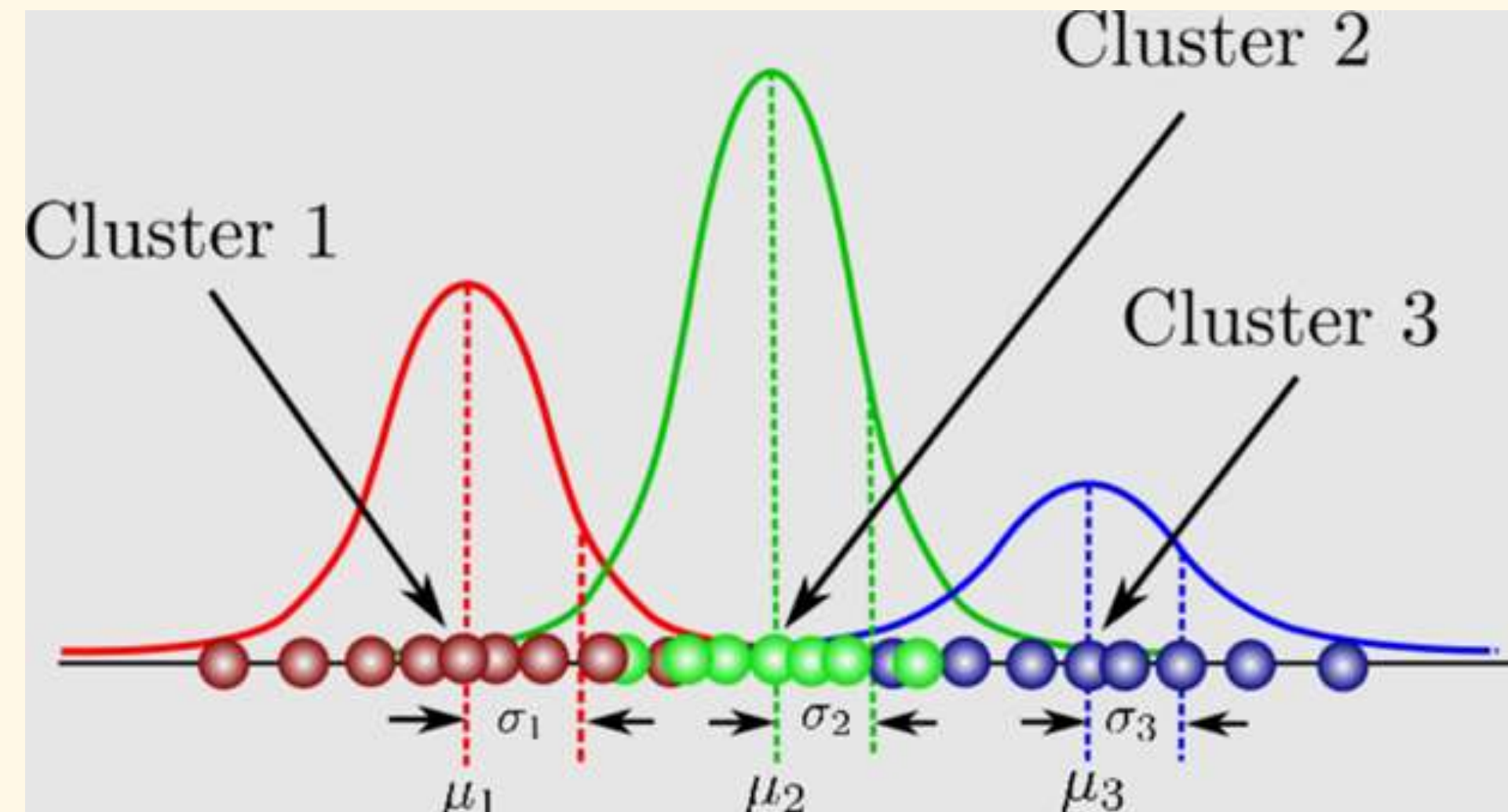
- ✓ Gaussian Mixture Models (GMM)

Strengths

- ✓ Flexible
- ✓ Provides probabilistic clustering

Limitations

- ✓ Computationally complex
- ✓ Assumes underlying distribution



Types of Clustering Techniques

We encode transactions into features:

Customer	Chicken	Burger	Fries	Sundae	Rice	Drink
C1	1	0	0	0	1	1
C2	0	1	1	0	0	1
C3	0	0	0	1	0	0
C4	1	0	0	0	1	0
C5	0	1	1	0	0	0
C6	0	0	0	1	0	0
C7	1	0	0	0	1	1
C8	0	1	0	0	0	1

1 = item purchased, 0 = not purchased

Types of Clustering Techniques

1. K-MEANS CLUSTERING EXAMPLE

Step 1: K = 3 clusters

Step 2: Expected Output Clusters

Cluster 1: “Meal Customers”

- ✓ C1, C4, C7
- ✓ Pattern: Chicken + Rice (+ Drink)

Cluster 2: “Snack Customers”

- ✓ C2, C5, C8
- ✓ Pattern: Burger + Fries + Drink

Cluster 3: “Dessert Customers”

- ✓ C3, C6
- ✓ Pattern: Sundae only

Customer	Chicken	Burger	Fries	Sundae	Rice	Drink
C1	1	0	0	0	1	1
C2	0	1	1	0	0	1
C3	0	0	0	1	0	0
C4	1	0	0	0	1	0
C5	0	1	1	0	0	0
C6	0	0	0	1	0	0
C7	1	0	0	0	1	1
C8	0	1	0	0	0	1

Interpretation:

- ✓ Clear behavioral segmentation
- ✓ Each cluster represents a **customer type**

Business Use

- ✓ Combo meals for Cluster 1
- ✓ Value meals for Cluster 2
- ✓ Dessert promos for Cluster 3

Types of Clustering Techniques

2. HIERARCHICAL CLUSTERING

EXAMPLE

Step 1: Start with Individual Customers

Each customer = its own cluster

Step 2: Merge Closest Pairs

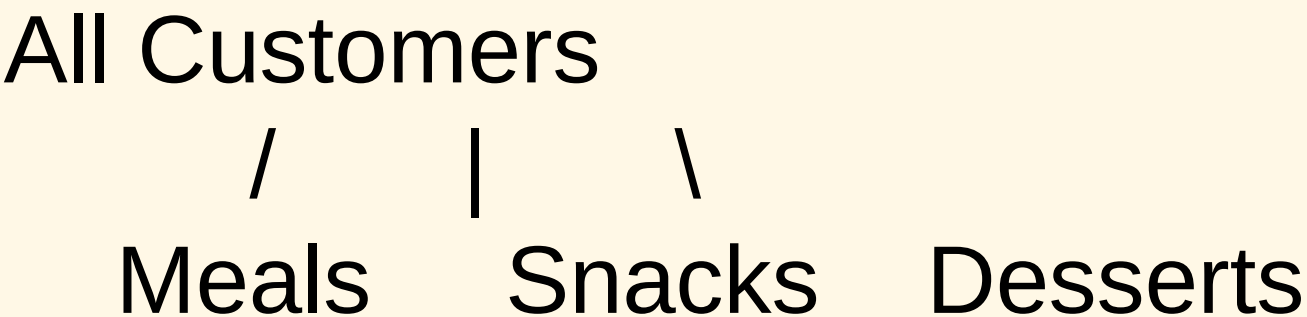
C3 + C6 → Dessert cluster

C1 + C4 + C7 → Meal cluster

C2 + C5 + C8 → Snack cluster

Step 3: Build Hierarchy

You get a structure like:



Customer	Chicken	Burger	Fries	Sundae	Rice	Drink
C1	1	0	0	0	1	1
C2	0	1	1	0	0	1
C3	0	0	0	1	0	0
C4	1	0	0	0	1	0
C5	0	1	1	0	0	0
C6	0	0	0	1	0	0
C7	1	0	0	0	1	1
C8	0	1	0	0	0	1

Shows **relationships between clusters**

You can decide:

2 clusters → (Meals vs Others)

3 clusters → (Meals, Snacks, Desserts)

Business Use

Flexible segmentation depending on strategy

Types of Clustering Techniques

3. DBSCAN (DENSITY-BASED)

EXAMPLE

- ✓ Dense groups = frequent behavior
- ✓ Sparse points = unusual behavior

Observations

Dense Regions:

- ✓ Chicken + Rice → common → Cluster
- ✓ Burger + Fries → common → Cluster

Sparse / Noise:

Sundae-only customers (C3, C6)

May be treated as:

- ✓ Small cluster OR
- ✓ Outliers (if rare enough)

Customer	Chicken	Burger	Fries	Sundae	Rice	Drink
C1	1	0	0	0	1	1
C2	0	1	1	0	0	1
C3	0	0	0	1	0	0
C4	1	0	0	0	1	0
C5	0	1	1	0	0	0
C6	0	0	0	1	0	0
C7	1	0	0	0	1	1
C8	0	1	0	0	0	1

Insight

DBSCAN may:

- ✓ Detect **core customer behaviors**
- ✓ Flag **rare ordering patterns**

Business Use

- ✓ Identify niche markets
- ✓ Detect unusual orders

Types of Clustering Techniques

4. MODEL-BASED CLUSTERING (GMM) EXAMPLE

Customers belong to clusters with **probabilities**

Example Output

Customer C8:

- ✓ 70% Snack cluster
- ✓ 30% Meal cluster

Why?

- ✓ Ordered Burger + Drink (but no fries)

Customer	Chicken	Burger	Fries	Sundae	Rice	Drink
C1	1	0	0	0	1	1
C2	0	1	1	0	0	1
C3	0	0	0	1	0	0
C4	1	0	0	0	1	0
C5	0	1	1	0	0	0
C6	0	0	0	1	0	0
C7	1	0	0	0	1	1
C8	0	1	0	0	0	1

Insight

- ✓ Not strictly one cluster
- ✓ Captures **real-world ambiguity**

Business Use

Personalized recommendations:

- ✓ Suggest fries to C8 (to complete snack pattern)

Types of Clustering Techniques

Technique	Result	Insight
K-Means	Clear groups	Best for simple segmentation
Hierarchical	Tree structure	Flexible grouping
DBSCAN	Dense clusters + noise	Finds anomalies
GMM	Probabilistic clusters	Handles overlap

How to Choose the “Best” Clustering Technique

1. If your goal is simple customer segmentation

Best choice: K-Means

- ✓ Fast and easy to implement
- ✓ Produces **clear, interpretable clusters**
- ✓ Works well for structured numeric data

Jollibee context:

- Segment customers into:
 - ✓ Meal buyers
 - ✓ Snack buyers
 - ✓ Dessert lovers
- ✓ Recommended for:
 - ✓ Dashboards
 - ✓ Business reporting

Types of Clustering Techniques

2. If you don't know how many clusters exist

Best choice: Hierarchical Clustering

Why:

- ✓ No need to define K initially
- ✓ Shows relationships via a tree (dendrogram)

Jollibee context:

Explore whether customers' form:

- ✓ 2 groups (Meals vs Others)
- ✓ 3 groups (Meals, Snacks, Desserts)

Recommended for:

- ✓ Exploration phase
- ✓ Research settings

3. If your data has noise or unusual behavior

Best choice: DBSCAN

Why:

- ✓ Detects **outliers automatically**
- ✓ Handles irregular cluster shapes

Jollibee context:

Identify:

- ✓ Rare orders (e.g., only gravy, only drink)
- ✓ Unusual customer patterns

Recommended for:

- ✓ Anomaly detection
- ✓ Real-world messy data

Types of Clustering Techniques

4. If customers show overlapping behavior

Best choice: Gaussian Mixture Model (GMM)

Why:

- ✓ Assigns **probabilities** instead of hard clusters
- ✓ More realistic for human behavior

Jollibee context:

A customer may be:

- ✓ 70% snack buyer
- ✓ 30% meal buyer

Recommended for:

- ✓ Personalization systems
- ✓ Recommendation engines

Situation	Best Technique
Fast, simple segmentation	K-Means
Unknown number of clusters	Hierarchical
Detect outliers	DBSCAN
Overlapping behavior	GMM

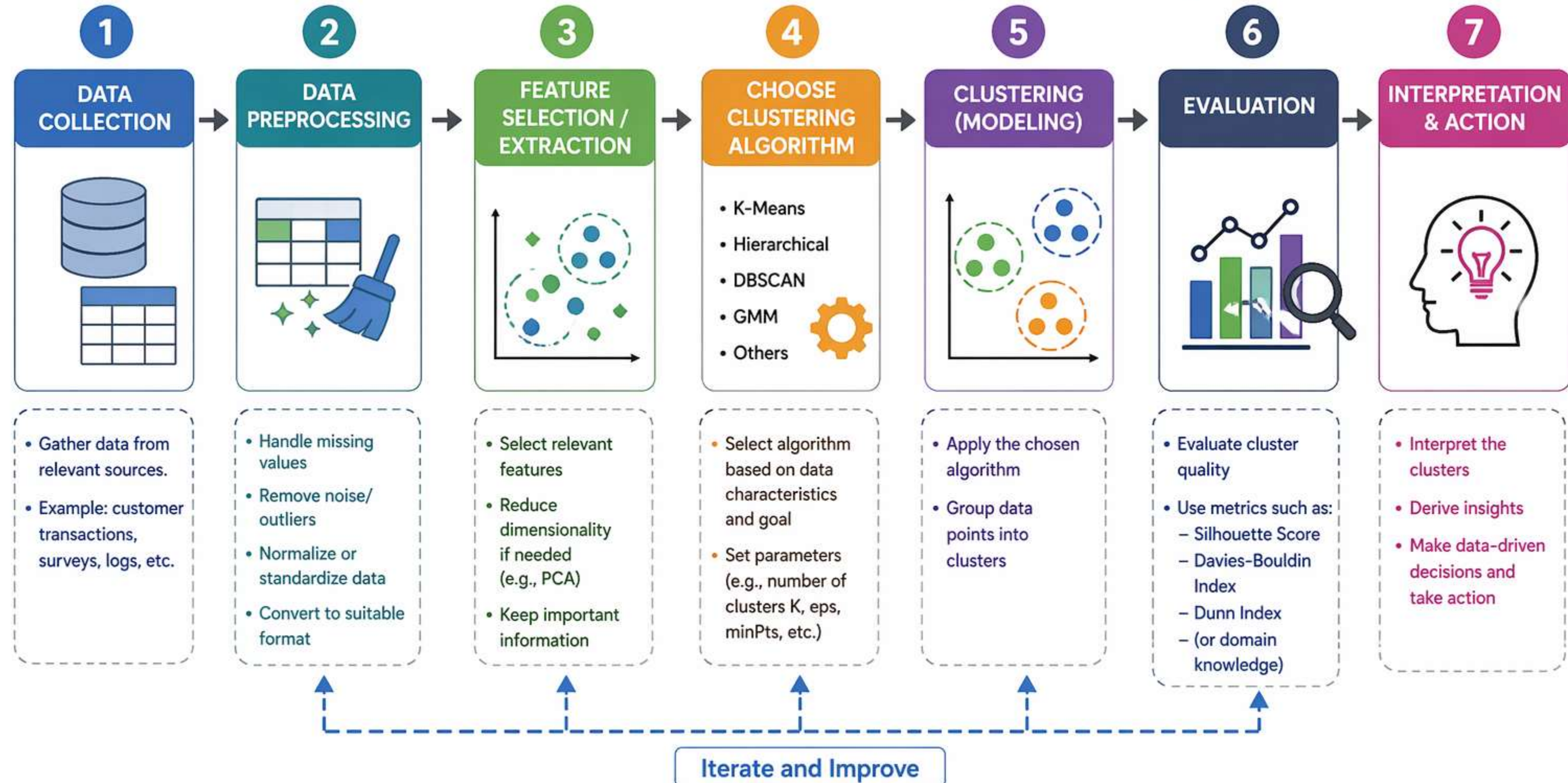
The “best” technique depends on:

- ✓ Data size
- ✓ Data type (numeric, text, binary)
- ✓ Noise level
- Interpretability needs

In practice:

Professionals often **try multiple methods and compare results.**

CLUSTERING PROCESS (PIPELINE)



NOTES



- Clustering is an iterative process. You may go back to earlier steps to improve results.
- There is no “one-size-fits-all” algorithm. The best choice depends on your data and objectives.

LEGEND (Example Cluster Results)



Cluster 1



Cluster 2



Cluster 3

EVALUATION OF CLUSTERING

Why Do We Need Evaluation?

Clustering is **unsupervised**, meaning:

- There are **no true labels**
- No direct “accuracy” like in classification

So we evaluate based on:

- ✓ **How well clusters are formed**
- ✓ **How meaningful they are**

Types of Clustering Evaluation

A. Internal Evaluation (Most Common)

- ✓ Uses only the dataset itself
- ✓ Measures structure of clusters

EVALUATION OF CLUSTERING

A1. Cohesion (Within-Cluster Similarity)

- Measures how **close data points are inside a cluster**

High cohesion = good clustering

Intuition:

Customers in the same cluster should behave similarly

Jollibee Example:

- ✓ Cluster = Meal Customers
- ✓ Chicken + Rice + Drink

Very similar orders → **high cohesion**

Bad case:

- ✓ Chicken + Sundae + Fries randomly mixed → Low cohesion

Cohesion Value	Meaning
Close to 0	Very tight cluster (GOOD)
Moderate	Acceptable
Large value	Spread-out cluster (BAD)

EVALUATION OF CLUSTERING

A2. Separation (Between-Cluster Difference)

- Measures how **distinct clusters are from each other**

High separation = better clustering

Example:

- ✓ Meal cluster vs Dessert cluster

Clearly different → **high separation** → Good case

Bad case:

- ✓ Meal cluster overlaps with snack cluster → Low separation

EVALUATION OF CLUSTERING

B. External Evaluation

✓ Used when ground truth labels exist

Compare clustering results to actual categories

Metrics:

- ✓ Purity
- ✓ Accuracy (rarely used)
- ✓ Adjusted Rand Index (ARI)

Example:

Suppose Jollibee already labels customers as:

- ✓ Meal buyer
- ✓ Snack buyer
- ✓ Dessert buyer

You compare:

- ✓ Your clusters vs actual labels

EVALUATION OF CLUSTERING

C. Relative Evaluation

Compare **different clustering results**

Example:

$K = 2$ vs $K = 3$ vs $K = 4$

Choose the best configuration

EVALUATION OF CLUSTERING

Common Evaluation Metrics

Silhouette Score

Formula Concept:

For each point:

- ✓ **a** = average distance within cluster
- ✓ **b** = distance to nearest other cluster

$$S = \frac{b - a}{\max(a, b)}$$

Range:

- **+1** → excellent clustering
- **0** → overlapping clusters
- **-1** → wrong clustering

Jollibee Example

Customer C1:

- ✓ Close to meal cluster → small **a**
 - ✓ Far from snack cluster → large **b**
- Silhouette ≈ **high (good)**

Bad case:

- ✓ Customer equally close to meal & snack
- Silhouette ≈ **0**

Score	Meaning
0.7 – 1.0	Strong clustering
0.5 – 0.7	Reasonable
0.25 – 0.5	Weak
< 0.25	Poor

EVALUATION OF CLUSTERING

Common Evaluation Metrics

Davies-Bouldin Index (DBI)

Measures:

- ✓ Cluster similarity
- ✓ Cluster compactness

Rule: Lower is better

Example:

- ✓ If meal & snack clusters overlap → DBI increases
- ✓ If clearly separated → DBI decreases

EVALUATION OF CLUSTERING

Common Evaluation Metrics

Dunn Index

Ratio of:

- ✓ Minimum inter-cluster distance to
- ✓ Maximum intra-cluster distance

Higher is better

Example:

Well-separated meal vs dessert clusters → high Dunn index

Overlapping clusters → low Dunn index

EVALUATION OF CLUSTERING

Common Evaluation Metrics

Within-Cluster Sum of Squares (WCSS)

- ✓ Used in **K-Means**
- ✓ Measures compactness
- ✓ Lower WCSS = tighter clusters

Example:

Customers in meal cluster very similar
low WCSS (good)

WCSS	Meaning
Low	Tight clusters (GOOD)
High	Spread clusters (BAD)

EVALUATION OF CLUSTERING

Common Evaluation Metrics

Elbow Method (for Choosing K)

Plot:

X-axis: number of clusters (K)

Y-axis: WCSS

Look for “elbow point”

Example:

- K=2 → high error
- K=3 → sharp drop
- K=4 → small improvement

Choose **K=3**

Practical Example (Jollibee Dataset)

Scenario:

You run K-Means with:

- K = 2, 3, 4

Results:

K	Silhouette	Interpretation
2	0.45	Too general
3	0.68	Good clustering
4	0.52	Over-segmentation

Best choice = **K = 3**

EVALUATION OF CLUSTERING

Domain Expert Evaluation

Domain expert evaluation is the process of **assessing clustering results using real-world knowledge** from subject matter experts.

- ✓ Instead of asking: “Are the clusters mathematically good?”
- ✓ We ask: “Do these clusters make sense in the real world?”

Mathematical metrics (Silhouette, WCSS, etc.) can say a cluster is “good,” but:

- ✗ It may not be interpretable
- ✗ It may not be useful
- ✗ It may not align with real behavior

Only domain experts can validate **practical relevance**

EVALUATION OF CLUSTERING

Who Are Domain Experts?

Depends on the problem:

- ✓ Business analysts
- ✓ Marketing managers
- ✓ Industry practitioners
- ✓ Subject specialists

Example: Jollibee

Domain experts could be:

- ✓ Store managers
- ✓ Marketing team
- ✓ Operations staff

EVALUATION OF CLUSTERING

Challenges in Clustering Evaluation

- **No Ground Truth.** Hard to validate objectively
- **Subjectivity.** Different algorithms → different results
- **Sensitivity.** Results depend on:
 - ✓ K
 - ✓ distance measure
 - ✓ scaling

EVALUATION OF CLUSTERING

Best Practices

- ✓ Use **multiple metrics**, not just one
- ✓ Combine **quantitative + qualitative analysis**
- ✓ Visualize clusters when possible
- ✓ Align with **business goals**

EVALUATION OF CLUSTERING

Business-Oriented Evaluation (Very Important)

In real systems like **Jollibee**, evaluation is not just mathematical.

Ask:

- Are clusters **actionable**?
- Do they support **marketing decisions**?
- Are they **interpretable**?

Example:

Cluster result:

- ✓ Cluster 1: Chicken + Rice
- ✓ Cluster 2: Burger + Fries
- ✓ Cluster 3: Sundae

Good clustering because:

- ✓ Easy to interpret
- ✓ Supports promotions
- ✓ Matches real behavior

EVALUATION OF CLUSTERING

Business-Oriented Evaluation (Very Important)

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Ask:

- Are clusters **actionable**?
- Do they support **marketing decisions**?
- Are they **interpretable**?

OUTLINE

- Clustering
- Topic Modeling



Pre-Activity

Scenario: “What Are Customers Really Saying?”

You are a data analyst at Jollibee.

You collected **thousands of customer comments**:

- “The chicken is crispy and delicious!”
- “Service is slow during lunch”
- “I love the spaghetti and burger combo”
- “The place is too crowded”
- “Affordable meals for students”

Pre-Activity

“If you had 10,000 comments... how would you summarize them?”

- ✓ Read manually ✗ (too slow)
- ✓ Categorize manually ✗
- ✓ Use clustering? ☐

“What if the computer can automatically discover **themes or topics**?”

TOPIC MODELING

Topic Modeling

Topic Modeling is an **unsupervised technique** used to discover **hidden themes (topics)** in a collection of documents.

Key Idea

Documents = mixture of topics

Topics = group of related words

Example (Jollibee Reviews)

Topic 1: chicken, crispy, juicy, delicious -> **Food Quality**

Topic 2: slow, service, waiting, staff -> **Service Issues**

Topic 3: cheap, affordable, budget -> **Pricing**

Topic Modeling

Difference: Clustering vs Topic Modeling

Aspect	Clustering	Topic Modeling
Output	Groups of documents	Topics (word distributions)
Focus	Similarity of items	Themes within text
Document	One cluster	Multiple topics

“The chicken is delicious but service is slow”

✓ Clustering → belongs to ONE group

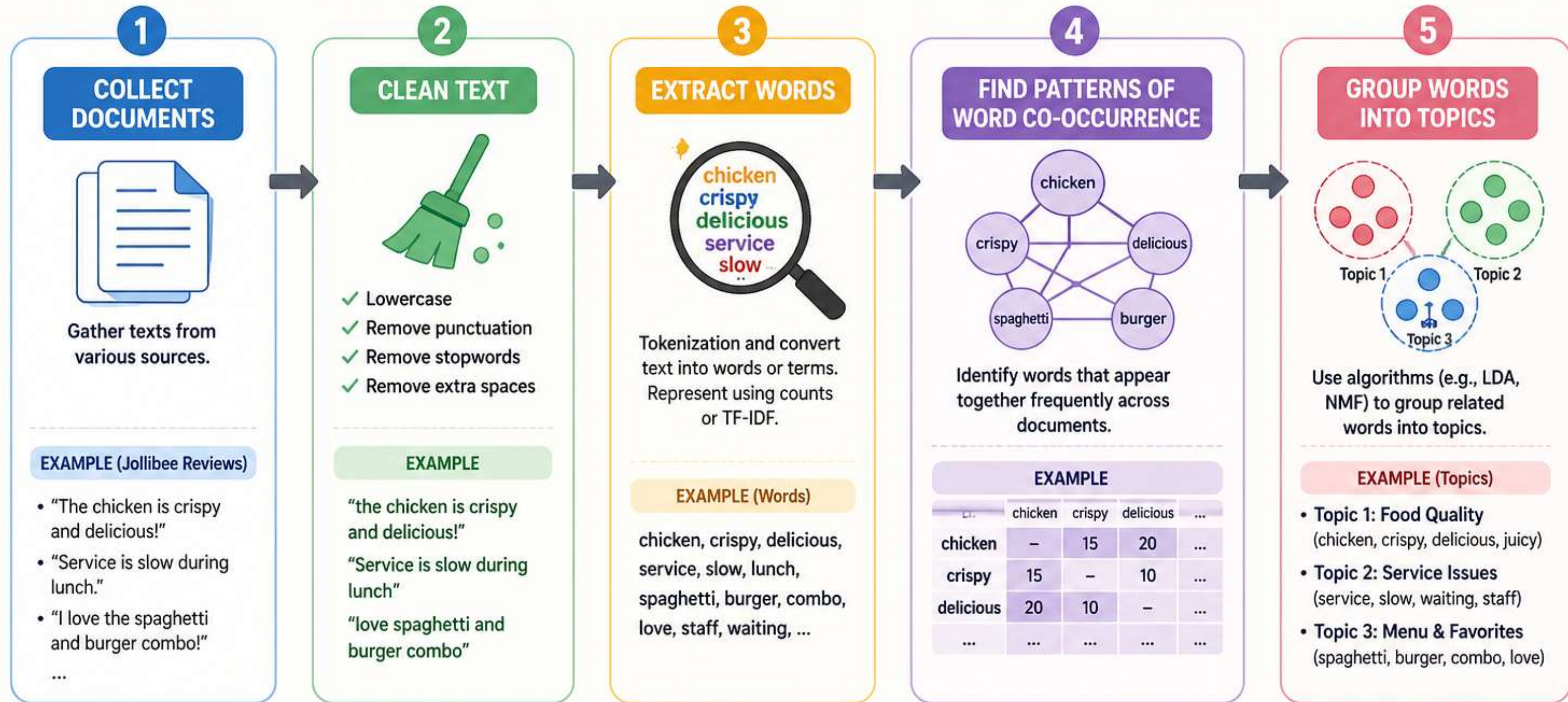
✓ Topic Modeling → belongs to:

- Food Quality
- Service Issues

More flexible!

TOPIC MODELING PIPELINE

From Raw Text to Meaningful Topics



WHAT IS TOPIC MODELING?

An unsupervised machine learning technique that discovers hidden themes (topics) in a collection of documents by finding groups of words that frequently occur together.

OUTPUT: INTERPRETABLE TOPICS



TOPIC 1 Food Quality

chicken, crispy,
delicious, juicy, tasty



TOPIC 2 Service Issues

service, slow, waiting,
staff, queue



TOPIC 3 Menu & Favorites

spaghetti, burger,
combo, love, favorite

These topics help understand what customers talk about most!

Words that appear together
frequently $\rightarrow$ form a topic

Common Techniques - LDA

A. Latent Dirichlet Allocation (LDA)

- generative statistical model that allows sets of observations to be explained by unobserved groups that explain why some parts of the data are similar.

Key Concepts:

- ✓ Document = mixture of topics
- ✓ Topic = probability distribution of words

Example

Topic 1: chicken (0.3), crispy (0.2), juicy (0.1)

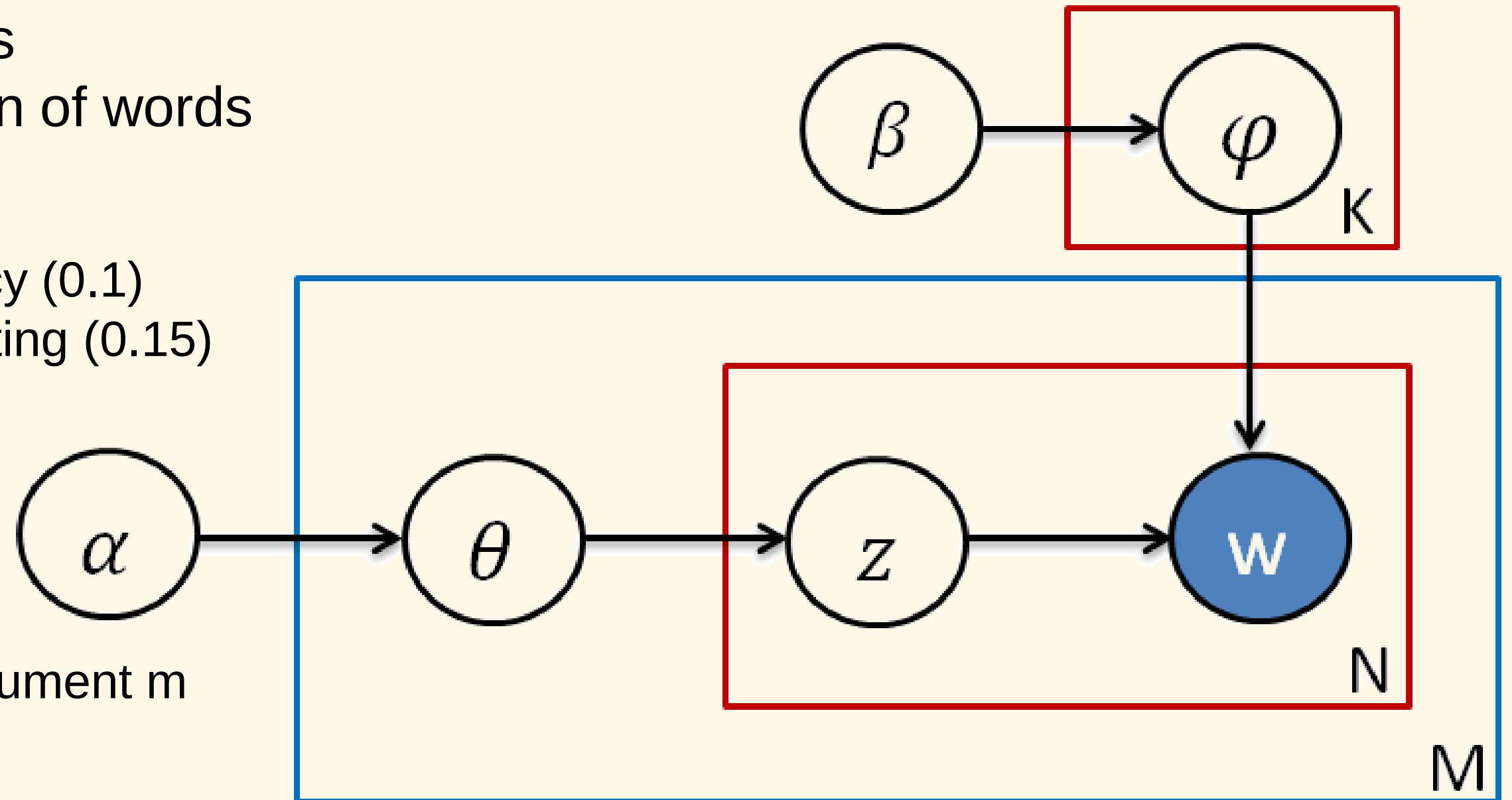
Topic 2: slow (0.25), service (0.2), waiting (0.15)

α - per-document topic distributions

β - per-topic word distribution

Θ - is the topic distribution from document m

Φ - is the word distribution



Common Techniques - LSA

B. Latent Semantic Analysis (LSA)

LSA is a topic modeling technique that:

- ✓ identifies **hidden (latent) relationships** between words and documents
- ✓ uses **matrix decomposition** to uncover patterns
- ✓ Words that appear in similar contexts have similar meanings.

1. How LSA Works (Simplified)
2. Build Term-Document Matrix
3. Apply Singular Value Decomposition (SVD)
4. Reduce dimensions
5. Extract hidden topics

Common Techniques - LSA

Example (Jollibee Reviews)

Input Documents:

- “Chicken is crispy and delicious”
- “Burger and fries are tasty”
- “Service is slow and staff is rude”

Term-Document Matrix (simplified):

Word	Doc1	Doc2	Doc3
chicken	1	0	0
burger	0	1	0
fries	0	1	0
service	0	0	1
slow	0	0	1

After LSA:

It may group:

- ✓ chicken, burger, fries → **Food Topic**
- ✓ service, slow → **Service Topic**

Common Techniques - LSA

Strengths of LSA

- Captures **synonyms** and relationships
- Reduces noise
- Works well with **smaller datasets**

Weaknesses of LSA

- Topics may be **hard to interpret**
- Allows **negative values** (less intuitive)
- Not probabilistic

When to Use LSA

- dataset is small
- you want dimensionality reduction + topic discovery

Common Techniques - NMF

C. Non-negative Matrix Factorization (NMF)

NMF is a topic modeling method that:

- factorizes a matrix into **non-negative components**
- produces **more interpretable topics**
- A document is made by combining parts (topics), and all parts are positive.

How NMF Works

1. Create TF-IDF matrix
2. Factorize into:
 - W (document-topic matrix)
 - H (topic-word matrix)

Common Techniques - NMF

Example (Jollibee Reviews)

Input:

“Chicken is crispy and delicious”

“Service is slow”

“Affordable meals for students”

Output Topics:

Topic 1: chicken, crispy, delicious

Topic 2: service, slow, waiting

Topic 3: affordable, cheap, budget

Common Techniques - NMF

Why NMF is Good

- No negative values
- Easier to interpret
- Cleaner topic separation

Strengths of NMF

- Highly interpretable
- Works well with TF-IDF
- Produces **clear topics**

Weaknesses

- Requires choosing number of topics
- Sensitive to initialization
- Not probabilistic (like LDA)

Common Techniques - NMF

When to Use NMF

- ✓ interpretability is important
- ✓ working with TF-IDF features
- ✓ analyzing survey responses

Common Techniques

LDA vs LSA vs NMF (Clear Comparison)

Feature	LDA	LSA	NMF
Type	Probabilistic	Linear algebra	Matrix factorization
Output	Probabilities	Latent structure	Additive components
Interpretability	High	Medium	High
Handles synonyms	Limited	Good	Moderate
Negative values	No	Yes	No

Common Techniques - Practical Comparison (Jollibee Case)

A. LDA

Produces: Food, Service, Price

✓ Easy to explain

B. LSA

Produces: Hidden relationships

✓ May mix topics

C. NMF

Produces: Clean, distinct topics

✓ Best balance of clarity and performance

Evaluation Topic Modeling

Why Evaluation in Topic Modeling is Challenging

Unlike clustering, topic modeling has **no ground truth labels**.

- ✓ You are not predicting a “correct answer.”
- ✓ You are **discovering structure**.

So evaluation must combine:

- ✓ Quantitative metrics
- ✓ Human interpretation
- ✓ Domain relevance

Evaluation Topic Modeling

Core Evaluation Dimensions

A good topic model should have:

- ✓ **Coherence**

Words in a topic make sense together

- ✓ **Distinctiveness**

Topics are clearly different

- ✓ **Interpretability**

Humans can label the topic

- ✓ **Stability**

Results are consistent across runs

Evaluation Topic Modeling

Quantitative Evaluation Metrics

A. Topic Coherence

Measures how semantically related the top words are.

Example (Jollibee)

- ✓ **High Coherence:** chicken, crispy, juicy, spicy (Clearly about food)
- ✗ **Low Coherence:** chicken, slow, table, price (Mixed meanings)

Score	Meaning
High	Good topics
Low	Random/unrelated

Evaluation Topic Modeling

Quantitative Evaluation Metrics

B. Perplexity (Mainly for LDA)

Measures how well the model predicts unseen data

Value	Meaning
Lower	Better statistical fit

Note that: Low perplexity $\neq$ good interpretability

Evaluation Topic Modeling

Quantitative Evaluation Metrics

C. Reconstruction Error (LSA/NMF)

How well the model reconstructs the original matrix

Value	Meaning
Lower	Better mathematical fit

May not reflect meaningful topics

Evaluation Topic Modeling

Quantitative Evaluation Metrics

D. Topic Diversity

Measures how different topics are from each other

Example

Poor diversity:

- ✓ Topic 1: chicken, crispy
- ✓ Topic 2: chicken, spicy

Too similar

Evaluation Topic Modeling

Qualitative Evaluation Metrics

A. Human Interpretability

Ask: Can we label this topic?

Does it make sense?

What label?

Example:

Topic 1: service, slow, staff, waiting

Label: **Service Issues**

✓ Good topic

Evaluation Topic Modeling

Qualitative Evaluation Metrics

B. Domain Expert Evaluation

Experts validate:

- ✓ relevance
- ✓ usefulness
- ✓ correctness

Example

Manager at Jollibee says:

- “Yes, service delays are a real issue”
- ✓ Model is useful

Evaluation Topic Modeling

Qualitative Evaluation Metrics

C. Intruder Test (Advanced but Powerful)

Procedure:

- ✓ Insert unrelated word into topic
- ✓ Ask users to identify the “intruder”

Example: chicken, crispy, juicy, laptop

“laptop” is the intruder

- ✓ Easy → good topic

Interpretation of Topics

Step 1: Look at Top Words

Example: chicken, crispy, juicy, spicy

Step 2: Assign Meaning

➤ “Food Quality”

Step 3: Validate with Documents

Check: Do documents in this topic match?

Step 4: Refine Label

Instead of: “Food”

Use: “Chicken Quality Experience”

Common Interpretation Problems

Mixed Topics: chicken, slow, price

Means:

- ✓ model is weak
- ✓ or K is incorrect

Too Generic: good, nice, okay

Not useful

Applications of Topic Modeling

A. Business (Jollibee)

- ✓ Customer feedback analysis
- ✓ Menu improvement
- ✓ Service optimization
- ✓ Marketing insights

B. Education (Your Context)

- ✓ Student feedback
- ✓ Survey analysis
- ✓ Program evaluation

C. Healthcare

- ✓ Patient feedback
- ✓ Complaint categorization

D. Social Media

- ✓ Trend detection
- ✓ Opinion mining

Applications of Topic Modeling

Output Topics:

Topic 1: tuition, free, help, family

➤ Financial Support

Topic 2: stress, requirements, deadline

➤ Academic Pressure

Topic 3: thankful, opportunity

➤ Positive Feedback

Evaluation:

- Coherence → High ✓
 - Distinct topics → Yes ✓
 - Interpretable → Yes ✓
- Model is good

Best Practices

- ✓ Combine metrics + human judgment
- ✓ Tune number of topics (K)
- ✓ Clean text properly
- ✓ Remove noise
- ✓ Validate with experts

“A good topic model is not just mathematically correct—it must be **meaningful, interpretable, and useful.**”

POST-EXERCISE:

- Do Machine Exercise: Clustering-Sy.ipynb
- Do Machine Exercise: Topic-Modeling-Sy.ipynb

Thank You!

Any Questions?